

Hsu, Chun-Yuan

 My Portfolio |  chyhsu@umich.edu |  Chyhsu |  Chun-Yuan Hsu

Skills

- Languages:** Python, C++, Go
- AI & ML:** PyTorch, LLM Integration (AWS Bedrock), RAG (ChromaDB), Embeddings, Semantic Search
- Cloud & DevOps:** Docker, Kubernetes, AWS, GCP, GitLab CI/CD, Grafana, NATS
- Databases:** MongoDB, Couchbase, ChromaDB, PostgreSQL, SQLite
- Frameworks/Systems:** Node.js, React, Linux (Debian, Arch), Git, Scrum

Education

Master of Science in Data Science , University of Michigan – Ann Arbor, MI, USA	Sep 2025–Present
Master of Science in Computer Science , National Tsing Hua University – Hsinchu, Taiwan	Sep 2022–Jan 2025
Bachelor of Science in Civil Engineering , National Cheng Kung University – Tainan, Taiwan	Sep 2018–Jun 2022

Experience

- Backend R&D Internship, QNAP, Taiwan** Jan 2025 – Jul 2025
- Built a retrieval-augmented Jira issue search system using AWS Bedrock and ChromaDB embeddings, increasing developer issue-resolution efficiency by 50%.
 - Developed an MCP-based Jira search server that integrates with IDEs, enabling developers to query and explore issues directly from their coding workflow.
 - Refactored Device Avatar microservices from Python to Go, achieving a 30% performance gain and optimizing deployment on Kubernetes.
 - Diagnosed and patched a critical memory leak in cloud production by correlating Grafana metrics with execution traces.

Projects

- Toward Interpretable Brain Age Prediction and AD Classification**  report
- Built an interpretable neuroimaging pipeline for brain age regression and Alzheimer’s Disease (AD) classification using structural MRI scans from OpenBHB and ADNI.
 - Adapted NeuroVFM, a pretrained 3D vision transformer, to extract patch-level MRI embeddings and aggregate variable-length features with attention-based multiple instance learning.
 - Trained joint classification and regression heads, reaching 0.873 diagnostic accuracy, 0.775 macro F1, 3.54-year MAE, and 0.966 R² while mapping attention back to clinically relevant brain regions.

Lilac

- Designed and implemented a cross-cloud IaC lifting framework that automatically reconstructs Terraform configurations from existing deployments (“brownfield” infrastructure) across Azure, Google Cloud, and AWS.
- Developed a pipeline that integrates Large Language Models (LLMs) with symbolic verification to learn resource–dependency mappings from cloud APIs.
- Conducted large-scale evaluation on real cloud environments, achieving higher accuracy and coverage than existing tools (e.g., Terraformer) while maintaining correctness.

- VizThinker** 
- Created an innovative graph-based interface for interacting with LLMs, reimagining traditional linear chatboxes into a visual conversation graph.
 - Implemented branching and node-based history navigation to support complex idea exploration.
 - Deployed on Google Cloud Platform with a modern stack of Node.js, React, and Python.

- OS-Nachos** 
- Implemented core OS concepts including system calls, multiprogramming, virtual memory, and file systems in C++.

- MIPS-CPU-Architecture** 
- Constructed a MIPS CPU architecture from the ground up using Verilog, implementing instruction fetching, decoding, and execution.